

Student Number

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PEM

2021

Year 11 End of Year Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using black pen
- NESA approved calculators may be used
- A NESA reference sheet is provided
- In Questions 13 – 30, show relevant mathematical reasoning and/or calculations

Total Marks – 80

Section I Questions 1 – 12 **12 marks**

Allow about 20 minutes for this section

Section II Questions 13 – 30 **68 marks**

Allow about 1 hour and 40 minutes for this section

Directions to School or College

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Section I (12 marks)

Write your answers on the multiple choice answer sheet.

Allow about 20 minutes for this section.

1 Which is the smallest volume?

- (A) 4000 mL
- (B) 0.365 L
- (C) 3.21 L
- (D) 375 mL

2 Which of the following sets of data contains an outlier?

- (A) 1, 2, 3, 4, 5, 5, 9
- (B) 1, 2, 3, 3, 4, 5, 7
- (C) 1, 2, 2, 3, 4, 6, 7
- (D) 3, 3, 3, 5, 5, 6, 11

3 Evaluate the following equation for d ,

$$d = \frac{V^2}{2\mu g}$$

by substituting $V = 60 \text{ km/h}$, $\mu = 0.8$ and $g = 9.8 \text{ m/s}^2$.

- (A) 229.59 m
- (B) 14112 m
- (C) 3.83 m
- (D) 0.0005 m

- 4 A nurse earns a yearly salary of \$56 200 before tax.
How much does she earn per fortnight?
- (A) \$1080.77
(B) \$4014.29
(C) \$2161.54
(D) \$5620.00
- 5 The movie “Penguin Bloom” starts at 10:40am and runs for 95 minutes.
The movie will end at
- (A) 1:05pm
(B) 1:40 pm
(C) 12:05pm
(D) 12:15pm
- 6 A protein bar has the following nutritional information.

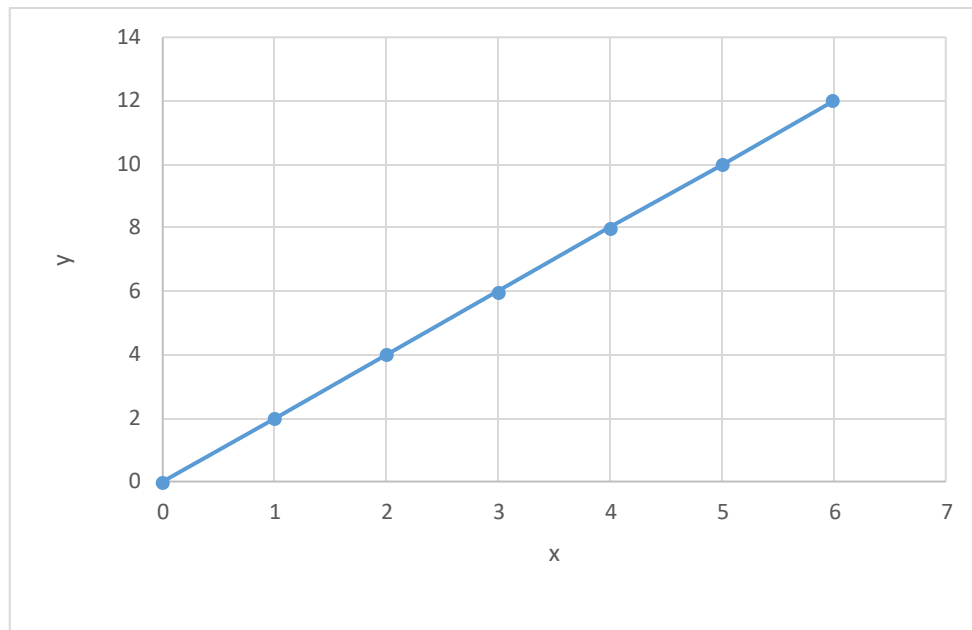
Servings per package: 1	Serving size 50g	
	Average Quantity per serve	Average Quantity per 100g
Energy	760kj	1520kj
Protein	15.8g	31.6g
Fat	6.9g	13.7g
Carbohydrate	2.4g	4.7g
Sodium	221mg	442mg
Dietary fibre	6.4g	12.8g

Joel has 2 protein bars a day and the recommended daily intake for an adult is 8700kj per day.

What percentage of the recommended daily intake does he have left for other food?

- (A) 91.26%
(B) 8.74%
(C) 17.47%
(D) 82.53%

- 7 The diagram below shows a straight-line graph.



The relationship between x and y is?

- (A) Direct, $y = 4x$
- (B) Direct, $y = 2x$
- (C) Not direct
- (D) Direct, $y = x$

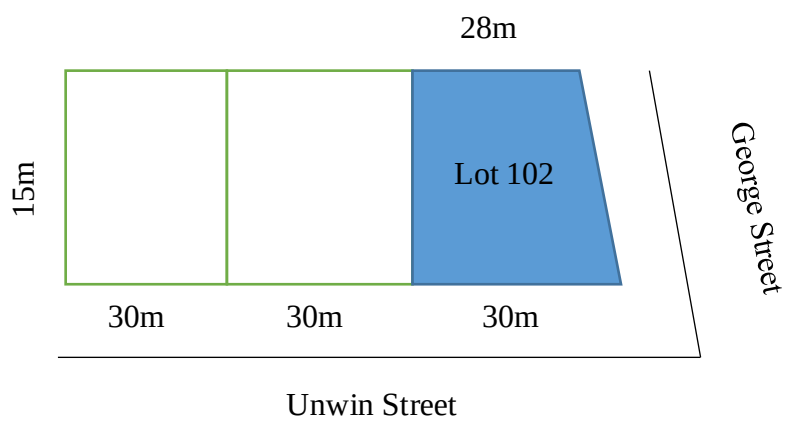
- 8 A set of Aeropods costs \$319.00 including 10% GST.



What is the price before GST was added?

- (A) \$350.90
- (B) \$287.10
- (C) \$305.90
- (D) \$290.00

9



Calculate the area of Lot 102 shaded in the diagram shown above.

- (A) 450 m^2
- (B) 435 m^2
- (C) 88 m^2
- (D) 420 m^2

- 10 Three university students go out to a bar.
Their blood alcohol content levels can be measured using the formulae

$$BAC_{\text{female}} = \frac{10N - 7.5H}{5.5M} \quad BAC_{\text{male}} = \frac{10N - 7.5H}{6.8M},$$

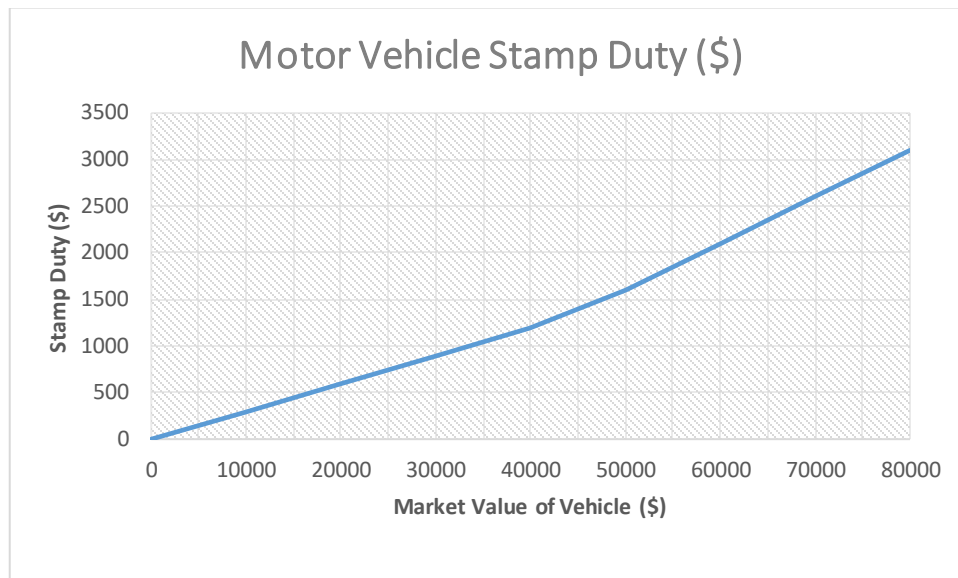
where N is the number of standard drinks consumed, H is the number of hours drinking and M is the person's weight in kilograms.

They each have three standard drinks in 2 hours. Mario weighs 85kg, Dylan weighs 105kg and Sarah weighs 70kg.

Which students will be under the limit of 0.05 BAC?

- (A) Only Sarah
- (B) Sarah and Dylan
- (C) All three students
- (D) Dylan and Mario

- 11 The following information is shown for the Stamp duty payable on New Motor Vehicles in NSW



Dylan buys a New Hyundai Tucson for \$52000.
Approximately how much will he pay in stamp duty?

- (A) \$1750
(B) \$2500
(C) \$2000
(D) \$2750
- 12 If two dice are tossed what is the probability that the sum of the uppermost faces is NEITHER 11 or 12.

- (A) $\frac{11}{12}$
(B) $\frac{1}{12}$
(C) $\frac{3}{12}$
(D) $\frac{2}{12}$

END OF SECTION I

Student Number

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Section II (68 marks)

Attempt Questions 13 - 30

Allow about 1 hour and 40 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided on pages **24** and **25**. If you use this space, clearly indicate which question you are answering.

Write your Student Number at the top of this page.

Please turn over

Question 13. (2 marks)

Solve the equation $7 - 2x = 3(x - 1)$.

2

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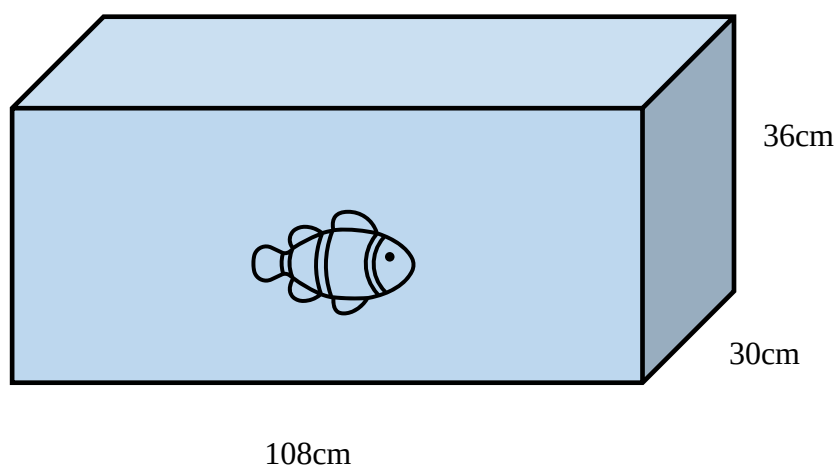
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Question 14. (3 marks)

The open fish tank below needs to have its glass replaced.



- a) Find the surface area of the open tank.

2

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- b) How much will it cost, to the nearest dollar, to replace the glass if the cost of glass is \$0.12 per square centimetre?

1

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Question 15. (3 marks)

The cost of hiring a pontoon boat on Lake Misty is \$70 flat plus \$30 per hour.

Joe and nine of his mates hire the pontoon boat for a 4-hour fishing trip.

- a) What is the total cost for the trip? **1**

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- b) How much does each person pay if they agree to share the total cost? **1**

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- c) Write a formula for the total cost (C) in dollars to hire a pontoon boat for (h) hours. **1**

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Question 16. (1 mark)

Perform the following calculation and round your answer to 2 significant figures. **1**

$$52.79 \times (2.3 \times 10^2)$$

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Question 17. (2 marks)

Clark's Formula calculates the Dose (D) of medication for a child using the weight of the child in kilograms (k), where A is the adult dose.

$$D = \frac{kA}{70}$$

Calculate the dosage in mg for a ten-year-old child weighing $30kg$ if the adult dosage is 400 mg .

2

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Question 18. (4 marks)

Consider the linear equation $y = 2x - 1$.

- a) State the gradient and y -intercept.

1

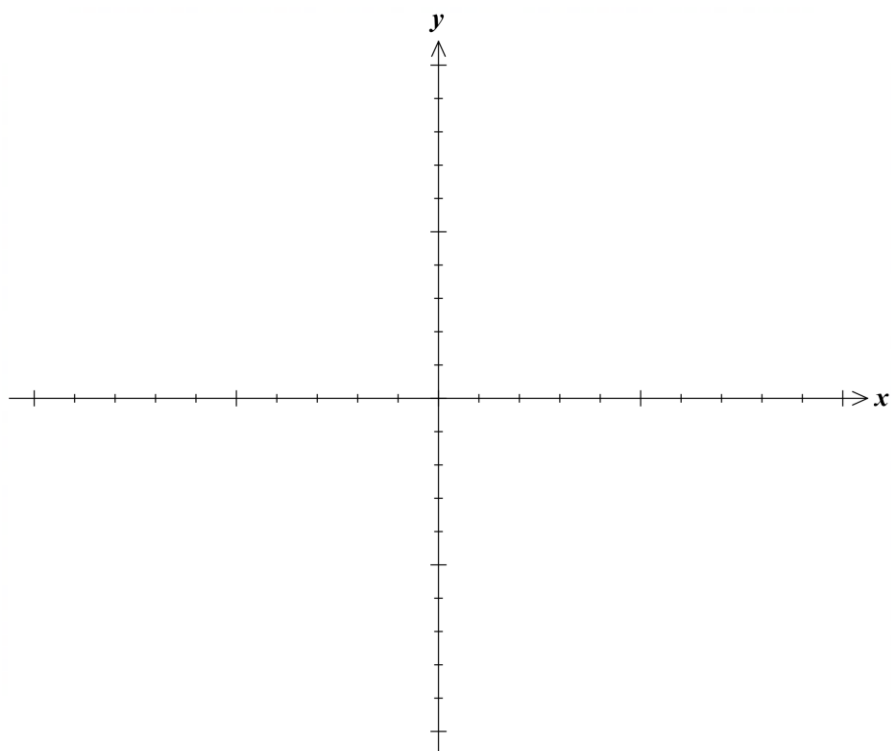
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- b) Sketch the graph of the function.

2



- c) Find the x -intercept.

1

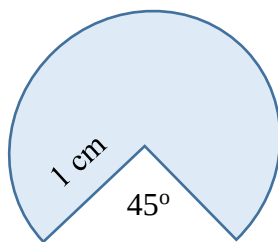
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Question 19. (4 marks)

Julia is making copper jewellery for her market stall. She decides to make 3 pendants, her design is shown below.



- a) Find the area of one pendant. Give your answer to two decimal places. 2

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- b) The dimensions of a sheet of copper are 7cm x 3cm. 2

If she fits 3 pendants on one sheet as shown



How much copper will be left over?

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Question 20. (4 marks)

Amy bought an 80 m^2 apartment off the plan, her contract states:

The builder may vary the size of the apartment by no more than 5%

When the apartment is built the total area is 77 m^2 .

- a) Is the apartment size within the boundaries of the contract? 2

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- b) The formula to convert square metres (SM) to squares (S) is given by 2

$$SM = 9.290304 \times S$$

Approximately how many squares is Amy's finished apartment?
(Give answer to 2 decimal places)

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Question 21. (4 marks)

Marta changed the password for her “Tweetie” account to a word with two digits on the end. Three weeks later when she went to access her account, she could remember the word but not the digits.

- a) What is the probability that she guesses the first digit correctly if all digits from 0 – 9 are possible? **1**

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- b) What is the probability that she guesses the first and second digits correctly? **1**

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- c) What is the probability that she guesses only one digit correctly? **2**

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Question 22. (3 marks)

The “Italiano” restaurant owners are charged a peak rate of 26.334 cents per kWh and an off-peak rate of 12.199 cents per kWh (for usage between 10 pm and 7am).

They use their 1800 W dishwasher for two and a half hours per day, six days a week, after 10pm when the restaurant closes.

How much will their quarterly (12 week) bill be for the dishwasher alone?

3

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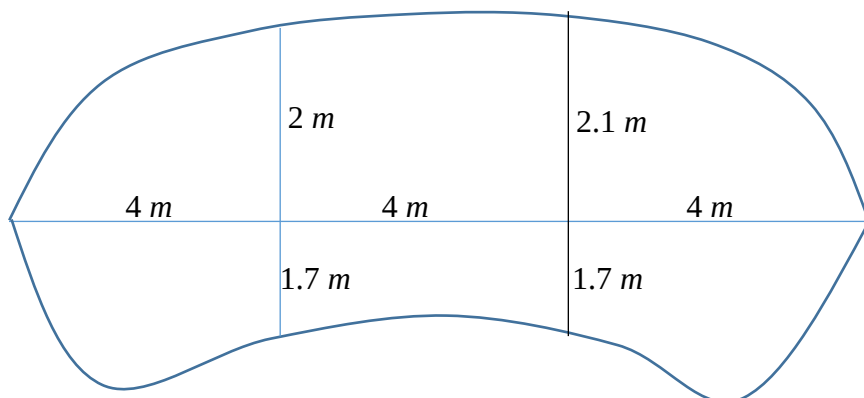
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Question 23. (5 marks)

- a) Use three applications of the Trapezoidal rule to estimate the area of the new “Tuscany” pool design introduced by Splash Pools and shown in the diagram below. 2



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- b) If the average pool depth is 1.2m, estimate the volume of water (in litres) needed to fill the pool. 2

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- c) Swimming pools require 40mL of liquid chlorine per 1000L of water. How much liquid chlorine will the “Tuscany” require? 1

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Question 24. (3 marks)

Consider the data presented below in a stem and leaf plot.

Stem	Leaf
0	0 1 2 5 6
1	1 2 3 3 4 5 9
2	0 1 4 4 5 5 7 8 8 8
3	3 4 5 6 8
4	1 2 4 6
5	8 8 9
6	2 6
7	0 7
8	5

- a) Comment on the shape of the distribution. **1**

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- b) Describe the modality of the data set? **1**

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- c) What is the range of the data? **1**

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Question 25. (4 marks)

A new 17-hour flight from London to Perth was introduced in 2019.

If the Frazer family leave London on the 9pm flight on Wednesday, the 7th of May.

- a) What day, date and time will it be in London when they arrive home to Perth? 2

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- b) What day, date and time will it be in Perth when they arrive? 2
(Perth is 8 hours ahead of London)

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Question 26. (5 marks)

All workers in Australia are paid superannuation. From 2017 the law requires employers to pay 9.5% of a workers' annual salary into a recognised superannuation fund.

Rod's company pays superannuation every quarter. Rod's salary is \$92500 for the 2020/2021 financial year.

- a) How much superannuation should his company deposit into his fund per quarter? **2**

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Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$45,000	19 cents for each \$1 over \$18,200
\$45,001 – \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

The above rates do not include the Medicare levy of 2%.

Rod has \$1200 in deductions for that financial year.

- b) How much will Rod pay in tax? **2**

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- c) Calculate Rod's Medicare levy. **1**

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Question 27. (4 marks)

A mortgage broker is paid an upfront commission and then a trail or ongoing commission for the life of the loan. The trail is calculated on the amount still owing at the end of each year of the loan.

Commission	Commission Rate
Upfront	0.65%
Trail	0.15%

- a) What is the upfront commission paid to the broker for a loan \$540 000? **1**

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- b) If, after one year, there is left owing \$528154, how much trail commission does the broker receive that year? **1**

.....

- c) If the broker signs up the following loans in a fortnight. How much is his upfront commission that fortnight? **2**

Loan amount	Term in years
\$250000	15
\$625000	30
\$120000	10

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Question 28. (7 marks)

A survey was undertaken of the number of years that the staff in the Loyal Bank City office have been employed.

The following list shows the number of years that the nineteen City staff have been employed:

1 2 3 4 4 5 6 7 8 9
11 12 15 16 18 22 24 25 35

- a) Determine the five-number summary for the city office staff data. 3

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- b) The distribution of the Country branch is shown in a box-and-whisker plot below. 2
Draw a second box-and-whisker plot to show the distribution of the City staff.



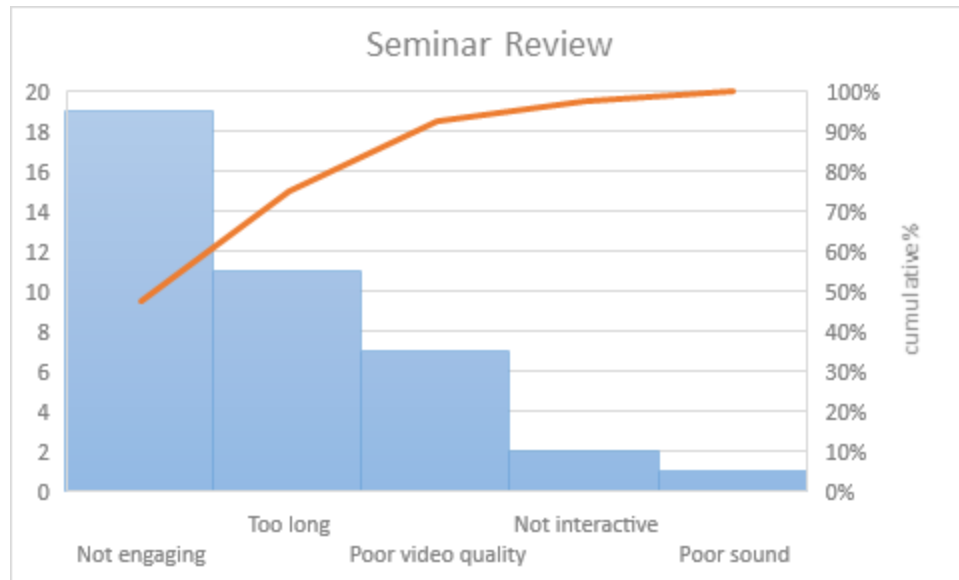
- c) Referring to the spread and skewness, comment on any similarities or differences between the two sets of data. 2

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Question 29. (3 marks)

An online seminar organiser asked participants to fill in a survey, the feedback was organised into the following Pareto chart.

a)



How many participants filled in the survey?

1

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b) Which feature(s) should the presenter work on to improve his presentation?

2

Give at least two ways he could improve.

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Question 30. (7 marks)

A sum of \$12000 is invested into an account when Kate turned 12 years old. It is to pay for her education at age 18. The savings account attracts 3% per year simple interest.

- a) Calculate the amount of interest earned each year. 1

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- b) Use the table below to calculate the amount the investment is worth over the 6 years. 2

Age	Amount in account
12	12000
13	12360
14	
15	
16	
17	
18	

How much will she have in the account at age 18?

.....

- c) At age 18 Kate buys a car to get to her TAFE course.

She pays a deposit of \$3000 from her account and her parents lend her \$10000 (for the balance and \$490 for new tyres). Kate agrees to pay them back \$80 per week.

- i) How long will it take to repay the loan from her parents? 2

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- ii) If she travels 20000km per year and a set of tyres will last 40000km. 2
Calculate the cost of tyre wear per kilometre.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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REFERENCE SHEET

Measurement

Precision

Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

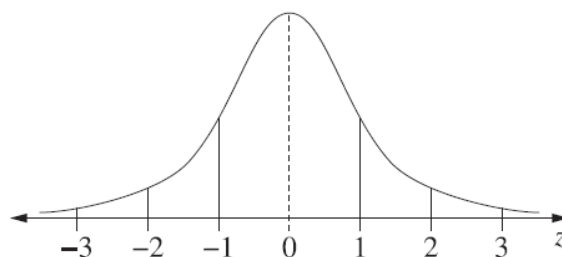
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

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MULTIPLE CHOICE ANSWER SHEET (12 marks)

Name:

Student Number:

Choose the best answer by filling in the circle corresponding to *A*, *B*, *C* or *D*.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
1	☐	☐	☐	☐
2	☐	☐	☐	☐
3	☐	☐	☐	☐
4	☐	☐	☐	☐
5	☐	☐	☐	☐
6	☐	☐	☐	☐
7	☐	☐	☐	☐
8	☐	☐	☐	☐
9	☐	☐	☐	☐
10	☐	☐	☐	☐
11	☐	☐	☐	☐
12	☐	☐	☐	☐